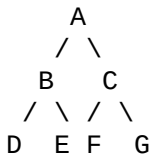
SIMATS ENGINEERING - Assessment Tool 3

Artificial Intelligence (CSA17) | CO2 | Annexure A: Dry Run Evaluation - Solutions

Question 1 - Breadth-First Search (BFS)



BFS uses a queue (FIFO); children are added left to right.

Iteration	Queue (after expanding)	Visited Node
1	B, C	A
2	C, D, E	B
3	D, E, F, G	C
4	E, F, G	D
5	F, G	E
6	G	F
7	(empty)	G

BFS traversal order: A, B, C, D, E, F, G

Question 2 - Depth-First Search (DFS)

DFS uses a stack (LIFO); the left child is explored first (top of stack shown first).

Iteration	Stack (top first, after expanding)	Visited Node
1	B, C	A
2	D, E, C	B
3	E, C	D
4	C	E
5	F, G	C
6	G	F
7	(empty)	G

DFS traversal order: A, B, D, E, C, F, G

Question 3 - 8-Puzzle using BFS

Initial: 136 / 5_2 / 478 **Goal:** 123 / 456 / 78_ (_ = blank, rows separated by /)

Moves are tried in the order Up, Down, Left, Right (direction of the blank); the move that undoes the previous one is skipped.

Iter	Current State	Queue Before Expansion	Queue After Expansion
1	S0 = 136/5_2/478	S0	S1, S2, S3, S4
2	S1 = 1_6/532/478	S1, S2, S3, S4	S2, S3, S4, S5, S6
3	S2 = 136/572/4_8	S2, S3, S4, S5, S6	S3, S4, S5, S6, S7, S8

Iter	Current State	Queue Before Expansion	Queue After Expansion
4	S3 = 136/_52/478	S3 - S8	S4 - S10
5	S4 = 136/52_/478	S4 - S10	S5 - S12

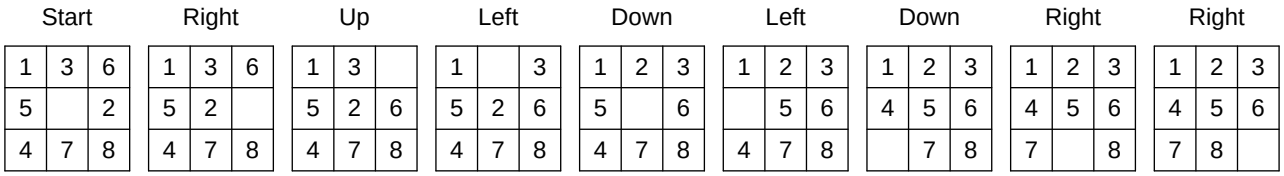
Generated states: S1 = Up, S2 = Down, S3 = Left, S4 = Right of S0. S5 = _16/532/478, S6 = 16_/532/478, S7 = 136/572/_48, S8 = 136/572/48_, S9 = _36/152/478, S10 = 136/452/_78, S11 = 13_/526/478, S12 = 136/528/47_.

Answers

1. Node expanded in each iteration: S0, S1, S2, S3, S4 (iterations 1 to 5). The goal lies at depth 8, so BFS continues beyond the 5 rows of the table.

2. States generated: 12 in the first 5 iterations. Until the goal is found, BFS expands all states at depths 0 to 7 and part of depth 8, i.e. a few hundred states; the exact count depends on duplicate checking and move order.

3. Solution path (8 moves): Right, Up, Left, Down, Left, Down, Right, Right.



4. Why BFS finds the shortest solution: BFS expands every node at depth d before any node at depth d+1. In the unweighted 8-puzzle every move costs 1, so path cost equals depth. The first time the goal is generated, it is at the smallest possible depth, so the path found is the shortest (optimal).